

# Translating Social Dilemmas into Gameful Designs for Wildfire Resilience

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Figure 1: Wildfire simulation games for Home Hardening and Community Evacuation

## Abstract

Developing wildfire resilience is a complex social problem since approaches to achieving it vary significantly across the population. Our interdisciplinary team designed serious games focusing on

dilemmas that affect decision-making for wildfire resilience and trade-offs among different decisions. We developed two serious games to aid individuals and communities in increasing their resilience towards wildfires, based on need finding interviews (N=21), which revealed three critical dilemmas around home hardening and evacuation. We contribute our collaborative reflection about the process, the design of the two games, and a small user study (N=5) of the games. Our findings suggest that our approach led to designing authentic, carefully balanced serious games to support community resilience through exposure to the dilemmas that occur

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in crisis situations and in preparing for such situations. The collaborative and interdisciplinary nature of our team allowed us to effectively translate social dilemmas into gameplay.

## CCS Concepts

• **Human-centered computing** → **Scenario-based design**.

## Keywords

serious games, wicked problems, dilemmas, community resilience, wildfire, evacuation, interdisciplinary collaboration

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## 1 Introduction

Developing serious games involves domain knowledge as well as game design practice [8], which becomes complex when the subject involves wicked problems [16] such as climate and natural hazard resilience. Wicked problems are impossible to fully formulate due to variations in perspectives in a pluralistic society on what is good or desirable [16]. In the case of climate and natural hazard resilience, there are many perspectives on which values to prioritize and how to weigh trade-offs. Therefore, we sought a strategy to uncover local values and to support meaningful communication and shared understanding toward more effective collective action. Through interdisciplinary brainstorming, collaborative reflection [14, 24] and need finding interviews (N=21), we built two serious games to engage the public with the complexity of wildfires using dilemmas and trade-offs. Together, our team bridged a social science understanding of human behavior and collective decision-making with game design and Human-Computer Interaction (HCI) to encourage behavior change through engaging interactivity. This resulted in games focused on collective action to help players reflect on trade-offs, decisions, and consequences of individual actions.

Our process included documenting the dilemmas faced by communities (identified through need finding interviews) and mapping those challenges to game mechanics, iterating within the researcher team, and developing and testing the games to gauge users' thoughts on the complex issues presented. Three main design components were selected through this process: multiplayer cooperative interactions to support discussion, narrative dialog to encourage empathy and consideration of other perspectives, and simulations to demonstrate the complex systems and interrelationships at play.

The primary contributions of this paper stem from collaborative reflection on the process of translating dilemmas into gameplay that deals with the challenging problem of wildfire resilience. This includes insights and implications that can be applied to other interdisciplinary projects addressing complex societal issues through serious games. We began by collaboratively critiquing the work of Johns et al. [7] – our games exist in the same domain, and we leverage insights from their community conversations and workshops.

However, [7] focus on presenting the player with information. Our collaborative reflection bridging serious games, social science, and HCI led us to present the player with dilemmas and difficult choices with no one right answer, toward sparking vital conversations and future action.

## 2 Related work

In this paper, we build on work related to social dilemmas and decision-making that occur in crisis situations like wildfire evacuations and various approaches to designing and developing wildfire interventions. This section synthesizes prior research and how our work builds on and is distinct from others.

*Social Dilemmas and Decision-Making.* Soden and Owen [19] consider individual actions within communities during a crisis or disaster and the informal tendency towards altruistic motives. Their work uncovers dilemmas faced by organizers and organizations in times of need, and while they focused on the COVID crisis, much can be learned from their approach and applied to dilemmas faced by individuals and communities in different crisis situations, such as wildfire. It is important to consider specific situations in which individuals may find themselves faced with a difficult choice. For example, a Red Flag Warning, issued when warm temperature, low humidity, and strong winds are expected to combine for an increased risk of fire danger [21], presents a dilemma to affected individuals: evacuate early in case there is a fire, or wait and see if a fire actually happens. Powell et al. [13] demonstrate the potential of co-designing games as a mechanism to improve understanding of wicked problems and spark action without forcing opinions to converge, and Nardo and Gaydos [10] discuss the value of incorporating wicked problems into games.

*Wildfire Risks and Interventions.* Scalingi [18] discusses the scope of ever-increasing challenges posed by large-scale wildfire events on humans and the environment and shares lessons learned from a series of such events that occurred across California, particularly noting the importance of ongoing collaborative actions that can be continuously implemented and adapted to local circumstances. Taking this to heart, we evolved our interactive approach as we communicated with different wildfire-prone communities and through our highly collaborative cross-disciplinary effort. In [12], attention is drawn to the importance of "incorporating local knowledge and social values" in building wildfire-resilient communities and landscapes. Similarly, our research suggests that individuals' values impact decision-making, and the trade-offs they personally face should not be overlooked. Where games are concerned, [7] uses participatory design to incorporate community ideas and values and ensure the games align with the specific needs of the target communities, while [3] utilizes a data-driven approach to design games that persuade based on real data. We build upon these approaches but take a unique angle by introducing dilemmas without obvious answers.

### 3 Design

#### 3.1 Need Finding

To better understand the factors influencing decision-making for wildfire preparedness and resilience among residents of high-fire-risk areas, we conducted a series of need-finding interviews with key stakeholders in two counties in California: Marin and Alameda. These counties were selected due to residential areas exposed to high wildfire risk as well as high levels of government, community, and individual action on wildfire risk reduction and preparedness.

Our interview sample included officials from state and local governments, such as regulators, city planners, and fire department representatives (n=8); professionals from the insurance industry—spanning insurance companies, research organizations, and agents (n=7); and members of community organizations, including Firesafe Councils and Firewise communities (n=6). Each interview involved a 60-minute conversation using Zoom. Interviewees were asked about key issues affecting household decisions to reduce wildfire risk and the implications of these decisions for individual and collective outcomes. Our interview protocol was approved by our university's Institutional Review Board.

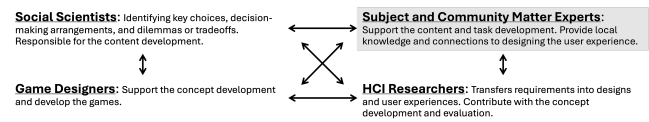
The interviewees highlighted different aspects of homeowners' knowledge, behavior, and decisions. The interviewees who work in state and local governments and community organizations were particularly concerned about the challenge of getting households to cooperate with the risk reduction guidance for defensible space and evacuation safety. Fire departments inspect individual lots for compliance with risk reduction requirements and distribute knowledge on mitigating risk and preparing for evacuation. However, officials and other community leaders worry that compliance is not sufficient to reduce risk at the community level, which could lead to faster wildfire spread and slower evacuation time. Interviewees in leadership roles in community organizations emphasized the challenge of convincing neighbors to cooperate with risk reduction planning and reported that even as peer communicators, it is hard to convince their neighbors to prepare for an uncertain risk. Many people choose to believe that "this will not happen to me" even though there is a higher and higher likelihood of wildfires developing in the area. Interviewees highlighted that residents face competing priorities, such as the cost of mitigation on private lots and the desire to save as much as they can during a fire, which could stress evacuation routes.

The interviews helped us understand that the key challenge is not in providing guidance based on authoritative knowledge. Rather, it is to help residents understand and manage the trade-offs among competing priorities that individuals face in confronting wildfire risk, as well as the impacts of individual decisions on the community. Inspired by insights from these interviews, we identified three key dilemmas that affect wildfire resilience in different stages of preparedness: investing in mitigation (home hardening), pre-evacuation on Red Flag days, and taking multiple cars when evacuating. These dilemmas expose misalignment between individual and collective interests in contexts of significant uncertainty. By introducing these dilemmas in the safe environment of the games, participants can understand the implications of different decisions through feedback from the game.

To analyze the need-finding interviews, we conducted thematic coding to identify key themes related to wildfire preparedness, community behavior, and decision-making challenges, ensuring alignment with real-world complexities while maintaining narrative authenticity.

#### 3.2 Interdisciplinary Team and Collaborative Reflection

Our team had eight members: two from social science, three HCI researchers, and three game designers. Subject matter and community experts were consulted as external collaborators but were not members of the core team; see Figure 2.



**Figure 2: Each member brought a unique skill set. Subject matter and community experts served as external consultants.**

We approached game development through iterative brainstorming and collaborative reflection. We conducted three hybrid brainstorming sessions combining Zoom and onsite discussions around scenarios and game mechanics and validated our decisions through interdisciplinary collaborative reflection. According to Prilla et al. [14], collaborative reflection is a method for teams to deliberate and discuss their approaches and decisions together either synchronously while being in a shared meeting or asynchronously, collecting and merging individuals' reflections into a collective understanding. The method has proven successful for other multidisciplinary, sensitive HCI design projects (e.g., [24]).

Wildfire resilience is a highly sensitive, complex subject, which can be traumatizing for users. Thus, it requires balancing emotional triggers while keeping the user experience relatable and understandable and simultaneously ensuring accuracy to the dangers posed to individuals living in fire-prone areas. We used collaborative reflection to iterate on our decision-making processes and identify any potential triggers and other challenges. For this, we identified central dilemmas and trade-offs faced by residents and designed gameplay and mechanics around each of those. Based on those decisions, we developed our prototypes and tested them with users to determine the effectiveness of our approach. Lastly, we also reflected on our overall approach and diverging yet complementary skills and backgrounds, highlighting benefits and open points in the interdisciplinary collaboration to develop wildfire-resilience games.

#### 3.3 Designing for Dilemmas

Our need-finding uncovered three dilemmas individuals face that could be translated into game scenarios and mechanics. Each dilemma involves at least three kinds of uncertainty: (1) uncertainty about the likelihood and timing of wildfire; (2) uncertainty about the actions of others in the community; (3) uncertainty about the impact of

actions on individual and collective outcomes. These uncertainties contribute to decision-making difficulties in resilience situations.

We developed a format for characterizing these social dilemmas as an input to game design. For each dilemma, we described the actors, the dilemma they face, and individual trade-offs among different decisions. These dilemmas were identified by examining uncertainties about wildfire risk, community behavior, and the consequences of individual actions. Through the analysis of main decision points, such as choosing between short-term personal safety and long-term community resilience, we translated these uncertainties into dilemmas that reveal conflicts between individual and collective interests. We then described four types of individual- and community-level decisions and implications: (1) the individual cooperates with public guidance, and most people in the community cooperate as well; (2) the individual cooperates, but most people in the community defect; (3) the individual defects but most community members cooperate; (4) the individual defects and most community members defect. Three dilemmas emerged as candidates, which we developed into game designs:

**Dilemma 1. Should I invest in mitigating wildfire risk on my property?** This dilemma occurs among residents who know they live in a high-fire-risk area but have limited information about if and when a wildfire will take place. Residents must balance the short-term costs of wildfire mitigation (e.g., clearing vegetation, using fire-resistant materials) against the long-term benefits of reducing individual and collective fire risk. Uncertainty about future wildfires, as well as the impact of such efforts on overall risk, complicates decision-making. Outcomes depend on collective behavior: If most property owners cooperate by mitigating risks, wildfire spread is less likely, benefiting the entire community. However, defectors—those who do not invest—may still benefit from reduced risk while their properties remain potential ignition points. If many residents defect, the collective fire risk remains high, resulting in greater losses for the entire neighborhood. This dilemma highlights the tension between short-term self-interest and the collective action needed to reduce wildfire risks effectively.

**Dilemma 2. Should I pre-evacuate on a Red Flag day?** This dilemma emerges when residents in high-fire-risk areas must decide whether to pre-evacuate during extreme fire weather, balancing personal inconvenience with collective safety. Pre-evacuation can reduce a community's evacuation time and save lives in case of a wildfire but comes at the price of disruption (e.g., missing work and discomfort) and financial cost, especially since not every extreme fire weather day will result in a wildfire. Outcomes vary based on collective behavior and fire progression: If no one pre-evacuates and a wildfire occurs, evacuation is slower, increasing fatalities and diverting emergency resources from firefighting to rescues. If many residents pre-evacuate, evacuation is faster, lives are saved, and emergency operations focus on containment. Cooperation requires individuals to sacrifice personal comfort for group benefit, but defectors—those who stay behind—avoid inconvenience if no wildfire develops. This underscores the tension between individual and group priorities, where the payoff for each resident depends on the collective response and whether the wildfire materializes.

**Dilemma 3. Should I evacuate with all my cars?** This dilemma arises during a wildfire when residents must evacuate immediately, facing a trade-off between personal advantage (saving multiple vehicles and possessions) and collective well-being (ensuring a swift evacuation for the community). High perceived risk and attachment to property complicate decisions, while the consequences of slower evacuation—threats to lives and diversion of emergency resources—may not be fully understood. Outcomes depend on collective behavior: If all residents evacuate with all vehicles, evacuation slows, increasing the risk of harm and straining emergency services. If residents evacuate in a small number of vehicles, evacuation is faster, but some possessions and vehicles are lost. Mixed behaviors lead to unequal outcomes: cooperators face losses, while defectors may benefit if they escape with their possessions, although slower evacuation risks dire consequences for all. The dilemma highlights the tension between individual priorities and collective safety during crisis scenarios.

| Choice                                                              | Individual cooperates (carpools)                                        | Individual defects (drives all cars)                                      |
|---------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Most people in the <b>community cooperate</b> by carpooling         | Fast evacuation - each participant gets to save one car                 | Fast evacuation. Defectors gain more than cooperators.                    |
| Most people in the <b>community defect</b> by driving multiple cars | Slower evacuation. Defectors gain more than cooperators if they escape. | Slow evacuation, dire consequences, diverts emergency response to rescue. |

**Table 1: Format for characterizing individual and community choices and outcomes in Dilemma 3**

Table 1, which describes Dilemma 3 regarding evacuation with all cars or one car, illustrates how we characterized outcomes for individual and community choices. This format aids game design by defining the roles of various actors and illustrating how their decisions affect both personal and community outcomes. Through scenarios like player-selected evacuation options, where in-game characters may not always follow instructions, the design process incorporates dilemmas and trade-offs to create engaging challenges. These mechanics aim to enhance the player's understanding of their own priorities and the broader community impacts of individual decisions.

### 3.4 Prototyping our Gameful Designs

To address these dilemmas in a gameful fashion, we employed a Research-through-Design process [5, 25] to prototype two minigames: one showing a top-down simulation of an evacuation with narratives for characters living in a community and the other a two-player game where two neighbors conduct home hardening and landscape fuel reduction tasks and a burn demo testing their decisions. The primary aim is to explore how these two approaches help support individuals and communities in wrestling with the dilemmas and trade-offs uncovered in our need-finding. Our games serve as proofs-of-concept and enabled us to run early user studies to test our designs.

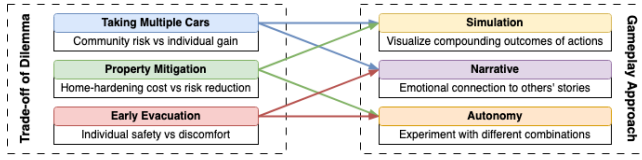


Figure 3: Mapping social dilemmas to playable systems

Inspired by entertainment games such as *Eco* [20], *House Flipper* [9], and *Animal Crossing* [11], alongside serious games research on motivation and autonomy [22], gameful design and pro-environmental attitudes [1, 4], and designing gameful systems with domain experts [2], we explored the following (and mapped to dilemmas in Figure 3):

- Narrative choices and character dialog help players uncover information about dilemmas faced by others [17]
- Simulated systems to portray wildfire and traffic challenges to non-expert users and show consequences [15]
- Autonomy and collaboration, i.e., allowing experimentation and discussion to support user discovery [23]

These game design decisions were chosen for their ability to shed light on the dilemmas of early evacuation, property mitigation, and taking one versus multiple cars. In particular, narrative elements encourage empathy about evacuation concerns, simulation aspects demonstrate compounding outcomes based on household choices, and autonomy and collaboration engage players with recognizing outcomes of their own decisions. These relationships are shown in the mapping in Figure 3.

### 3.5 Game Concepts

Based on the dilemmas and our exploration of narrative, simulation, autonomy, and multiplayer interactions, we developed two games: *Firesafe Friends*, which allows two players to perform home-hardening, and *Firewise Citizens*, which shows a top-down simulation of a fire and evacuation. The games provide key information through text and visual feedback and animations (e.g., visualization of cars on the road, with more cars causing traffic jams if two-car households take both cars, and slower evacuation if everyone waits until the notice instead of pre-evacuating). A news report at the conclusion of the game helps contextualize the results of different decisions to allow players to understand the outcomes and consequences. In addition to being informative, this news report encourages empathy by showing the player the difficult choices faced by some residents and the impacts their decisions can have on the whole community.

As seen in Fig 4, in the *Firesafe Friends* game, the user interacts with a house by selecting and purchasing materials (with a limited budget) and applying those materials to "harden" the home, making it more resistant to embers and flames. In the second game, *Firewise Citizens*, the user interacts by zooming in on individual houses and talking to the residents about their individual situations, and suggesting actions to take before or during an evacuation. Both games are available online at <https://ucsc-wildfire-games.itch.io/firesafe-friends-prototype> and <https://ucsc-wildfire-games.itch.io/firewise-citizens>.



Figure 4: (Top) *Firesafe Friends* UI for home hardening, (Bottom) *Firewise Citizens* UI for conversations with residents

## 4 User Study

To explore the potential impact of our designs, we conducted a preliminary user study focused on understanding participants' engagement with our game prototypes and any changes in thought about wildfire preparedness that resulted from playing. We recruited five participants for this study during a game-focused conference, selected for their general knowledge of game design and ability to play a game in a prototype state. Three played *Firewise Citizens*, one played *Firesafe Friends*, and one played both. Two participants were aged between 18 and 24, two between 24 and 35, and one between 35 and 44. Two participants identified as male, one as female, and two as non-binary/third gender. All participants were asked to consent before beginning the study and were given the option to quit at any time. The study involved participants playing the game(s) and completing surveys to capture their feelings about preparedness, gameplay experiences, and suggestions for improvement.

During the pre-survey, participants were asked about their preparedness for wildfires, likelihood of following evacuation orders, and willingness to help others during a disaster. These questions were designed to establish baseline attitudes and behaviors. Participants then played one of two games: *Firewise Citizens* or *Firesafe Friends* (one player played both games), designed to expose players to wildfire dilemmas and decision-making. In the gameplay phase, participants engaged in a think-aloud session [6], providing insights into their decision-making processes. Finally, a post-survey gathered feedback on player sentiment and suggestions for improvement.

#### 4.1 Early Findings from Preliminary User Study

A light qualitative analysis of participant responses revealed three main themes:

**Understanding Situations and Perspectives:** The exploration of realistic scenarios and decision-making was engaging for participants. *P3: "The game added perspective to the position emergency personnel are put in when advising civilians."* and *P2: "Holistic alignment on why residents may or may not comply, and their concerns."*

**Developing Awareness Through Gameplay:** Participants noted that the games helped them better understand wildfire preparedness. *P1: "Learning about different evaluation [sic] and prevention options through dialogue was eye-opening."* and *P2: "Understanding that every component of a house matters when it comes to wildfire safety."*

**Motivation for Preparedness:** Most participants expressed increased motivation to learn about preparedness after playing. *P5: "Now I want to learn more about home hardening."* and *P6: "It motivated me to think about small changes renters can make to improve safety."*

## 5 Discussion and Next Steps

Collaborative reflection on the games and early user study results led to the insight that translating dilemmas into game scenarios requires a focus on local and practical specifics, such as the main mode of transportation or average number of cars per household, and less focus on a theoretical understanding of wildfire resilience. It is premature to attempt in this paper to assess the feedback between games and community actions. Open research questions resulting from the collaboration will inform this assessment, including what measurements to develop to evaluate the impact of the gaming experience on community knowledge of dilemmas and how to assess any increase in preparedness plans and actions.

Our observation of past works such as [7] indicated an inclination towards *instructing* players and guiding them towards *right answers*. With our approach, we aimed to center dilemmas and trade-offs in the gameplay while still maintaining the important aspect of including accurate information and realistic scenarios. This allowed us to shed light on the difficult situations people face and to encourage the player to explore these scenarios through narrative, simulation, autonomous experimentation, and multiplayer collaboration and discussion.

Our serious games enhance wildfire preparedness by helping residents understand dilemmas so they can have more informed community discussions toward collective actions. This paper represents ongoing work, and we share these early findings with the HCI community to encourage discussion about the importance of meaningful interdisciplinary collaboration on projects that address wicked problems and the value of introducing trade-offs and dilemmas to engage with the public about such issues. For our team, this approach enabled us to maintain a shared vision and find connection points across disciplines and allowed us to respectfully and constructively communicate our ideas.

#### 5.1 Translating Social Dilemmas into Digital Games

What emerged from our collaborative efforts in designing a gameful wildfire intervention is a clear need to improve user understanding of dilemmas and trade-offs in crisis decision-making situations and to convey the impact of their decisions on others. To accomplish this, we identified aspects of gameplay that support player exploration of different situational concerns (narrative), showing compounding results of decisions (simulation), and allowing the user freedom to experiment with different combinations of options without direct guidance (autonomy). Future work will explore the benefits of each strategy in a more fine-grained manner.

#### 5.2 Implications for Future Interdisciplinary Research Collaborations

Based on the reflections of our team, we offer process considerations for discussion with other researchers and designers working at this intersection of serious games and complex societal problems:

- Collaborating across disciplines to approach need finding & social dilemma development: having multiple disciplines represented within the same team as a unified force with a collective vision enabled better engagement with cross-disciplinary thoughts and ideas.
- Transferring social dilemmas into interactive game designs and the role of interdisciplinary collaboration: we needed to consult each other about the format of information sharing early and often to stay on the same page.
- Benefits of collaborative reflection: having everyone together in-person playing the games and discussing the fit between the games and the dilemmas enhanced our ability to iterate the design carefully and mitigate potential negative impacts, such as triggering traumatic memories of past experiences.

Our study was limited by the small sample size, and future studies with these or other games should ensure a broad sample of the population to assess the effectiveness of the games and of the process we took incorporating dilemmas and trade-offs and understanding the games through interdisciplinary perspectives.

## 6 Conclusion

In this paper, we shared the early results of our interdisciplinary and reflective process of designing two serious games centered on helping individuals grapple with dilemmas and trade-offs in wildfire resilience. We share our findings from collaborative reflection as well as insights from interviews and a preliminary user study. These results may benefit researchers, designers, and the broader HCI community engaged in addressing complex social challenges, particularly in areas such as disaster preparedness, climate resilience, and community-driven decision-making. Our findings provide insights for those developing interactive systems and serious games aimed at fostering collective action and behavior change in uncertain environments.

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## A Pre-survey

How old are you?

- 18-24 years old
- 25-34 years old
- 35-44 years old
- 45-54 years old
- 55-64 years old
- 65+ years old
- Prefer not to say

How do you describe yourself?

- Male
- Female
- Non-binary / third gender
- Prefer to self-describe
- Prefer not to say

How many children under 18 live with you?

Are you a Homeowner or Renter?

- Homeowner
- Renter
- Other

Do you live in the U.S.?

Do you believe you live in an area impacted by wildfire?

- Definitely not
- Probably not
- Might or might not
- Probably yes
- Definitely yes

Do you believe you (or your family) are prepared for a wildfire?

- Definitely not
- Probably not
- Might or might not
- Probably yes
- Definitely yes

Do you worry about a wildfire or evacuation?

- Often
- Sometimes
- Rarely
- Never

Have you experienced a wildfire or other type of evacuation before?

- Experienced a Wildfire
- Experienced other type of evacuation
- Experienced neither

How likely are you to follow instructions for an evacuation order during a wildfire or other disaster?

- Very likely
- Somewhat likely
- Very unlikely
- How likely are you to help others during a wildfire or other type of disaster?
- Very likely
- Somewhat likely
- Very unlikely

## B Post-survey

Do you believe you (or your family) are prepared for a wildfire?

- Definitely not
- Probably not
- Might or might not
- Probably yes
- Definitely yes

How likely are you to follow instructions for an evacuation order during a wildfire or other disaster?

- Very likely
- Somewhat likely
- Very unlikely

How likely are you to help others during a wildfire or other type of disaster?

- Very likely
- Somewhat likely
- Very unlikely

Which game did you play?

- Firesafe Friends
- Firesafe Citizens
- Happy House
- Info Overload
- Find Your Things
- Alert the City

What aspects of the game do you think helps most with wildfire preparedness? (open response)

Do you have any suggestions for things to add, change, or remove? (open response)

If the game had the possibility for negative outcomes, such as loss of life during the fire, would that impact your willingness to play again?

- I would be more likely to play again to try to get a better outcome
- I would be discouraged from playing again
- It would not impact my willingness to play again

Who do you think this game is for, or who do you think would benefit the most from playing? (open response)

Is there anyone you would like to show this to, and why? (open response)

What key takeaways or new ideas do you have after playing? (open response)

How motivated are you to find additional information about preparing for wildfire? Did the game impact your motivation? (open response)

Did you have a conversation with anyone while playing?

- No conversation
- Minimal conversation
- Some conversation
- Lots of conversation

Did you play with someone else?

- Yes
- No

How did you know them before playing?

- Not at all
- Somewhat
- Very well

How well did you work together?

- Terrible
- Not well
- Pretty well
- Excellent

Rate the following for the sentence "The game was \_"  
Enjoyable:

- Not at all enjoyable
- Not very enjoyable
- Somewhat enjoyable
- Very enjoyable

Challenging:

- Not at all challenging
- Not very challenging
- Somewhat challenging
- Very challenging

Informative:

- Not at all informative
- Not very informative
- Somewhat informative
- Very informative

Persuasive:

- Not at all persuasive
- Not very persuasive

- Somewhat persuasive
- Very persuasive

Frustrating:

- Not at all frustrating
- Not very frustrating
- Somewhat frustrating

- Very frustrating

Thought Provoking:

- Not at all thought provoking
- Not very thought provoking
- Somewhat thought provoking
- Very thought provoking